

EXPLORATORY DATA ANALYSIS

Gaming Behavior and Academic Performance

Tuwaiq Academy AI Bootcamp Project

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Dataset Understanding

What is the dataset about?

Students' gaming behavior, lifestyle habits and academic performance — gaming, study, sleep, attendance, device use, stress, addiction score and grades.

What does a row represent?

One student, with their gaming, lifestyle and academic information.

How many rows and columns?

Original $8,000 \times 14$. Working set, scope-restricted to Male/Female: 7,707 rows.

Numerical vs. categorical?

10 numerical features, 3 categorical (gender, gaming_genre, stress_level) and 1 identifier (student_id).

What is the target?

grades — a continuous numerical variable, so a regression-style problem.

Initial observations?

Well structured, no missing values or duplicates — but dtype fixes were needed, some grades exceeded 0–100, some addiction scores were negative, and two scales are undocumented.

Data Dictionary

COLUMN	DESCRIPTION	UNIT / SCALE	ROLE
student_id	Unique student identifier	integer ID	Identifier
age	Student age	years	Feature
gender	Reported gender	Male / Female	Feature (categorical)
gaming_hours	Time spent gaming per day	hours	Feature
study_hours	Time spent studying per day	hours	Feature
sleep_hours	Sleep per night	hours	Feature
attendance	Class attendance rate	percent	Feature
social_activity	Social activity level	index	Feature
device_usage	Daily screen time across devices	hours	Feature
reaction_time_ms	Reaction-test latency	milliseconds	Feature
addiction_score	Gaming addiction indicator	undocumented scale	Feature
gaming_genre	Preferred genre	FPS / Casual / RPG	Feature (categorical)
stress_level	Self-reported stress	Low / Medium / High	Feature (categorical)
grades	Academic performance score	0–100	Target

Data Quality Assessment

CHECK	FINDING	DECISION
Missing values	0 across all 14 columns	No treatment required
Duplicates	0 — exact, without-ID and ID-only checks	All rows kept
Invalid values	130 grades outside 0–100; 103 negative addiction scores	Offending cells flagged and set to NaN — not capped, not dropped
Statistical outliers (IQR)	0 detected in any numerical column	Invalid ≠ outlier — range checks caught what IQR could not
Categorical consistency	No hidden case or whitespace variants	Categories cast to category dtype

Summary Statistics

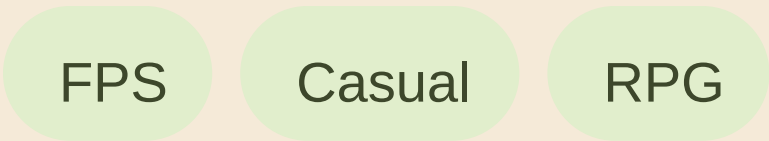
Valid grades only — after the invalid cells were nulled.



GENDER



GAMING_GENRE



STRESS_LEVEL

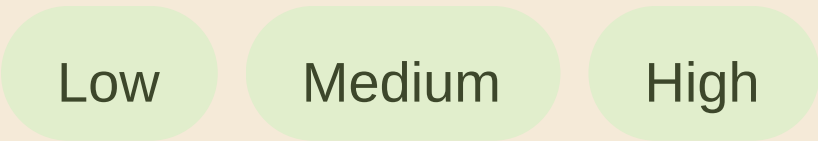
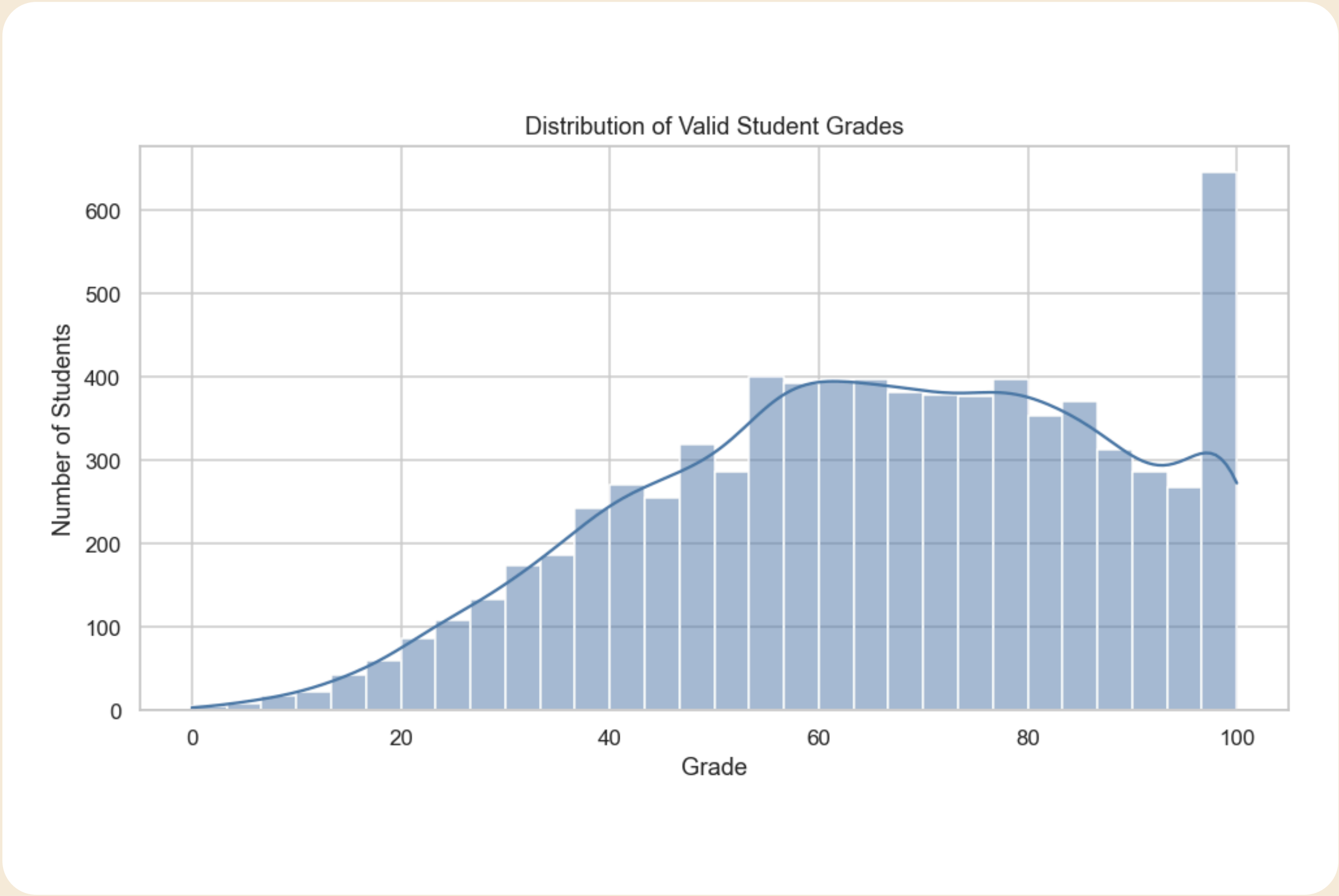


FIGURE 1

Distribution of Valid Grades



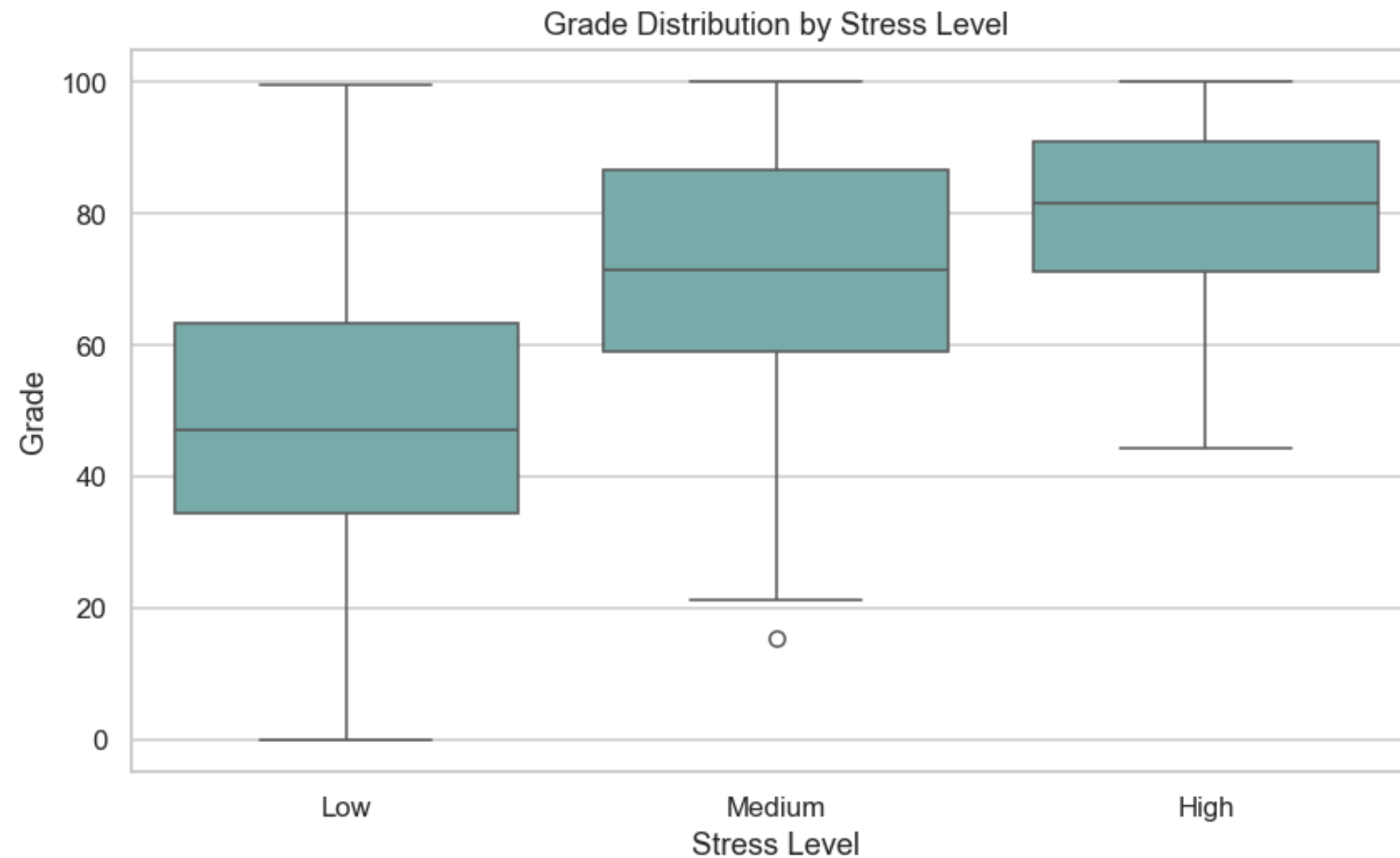
0–100

FULL RANGE USED

Grades span the entire scale and sit broadly around the mid-60s, with no severe skew and no pile-up at either bound — healthy variation across students.

FIGURE 2

Grades by Stress Level



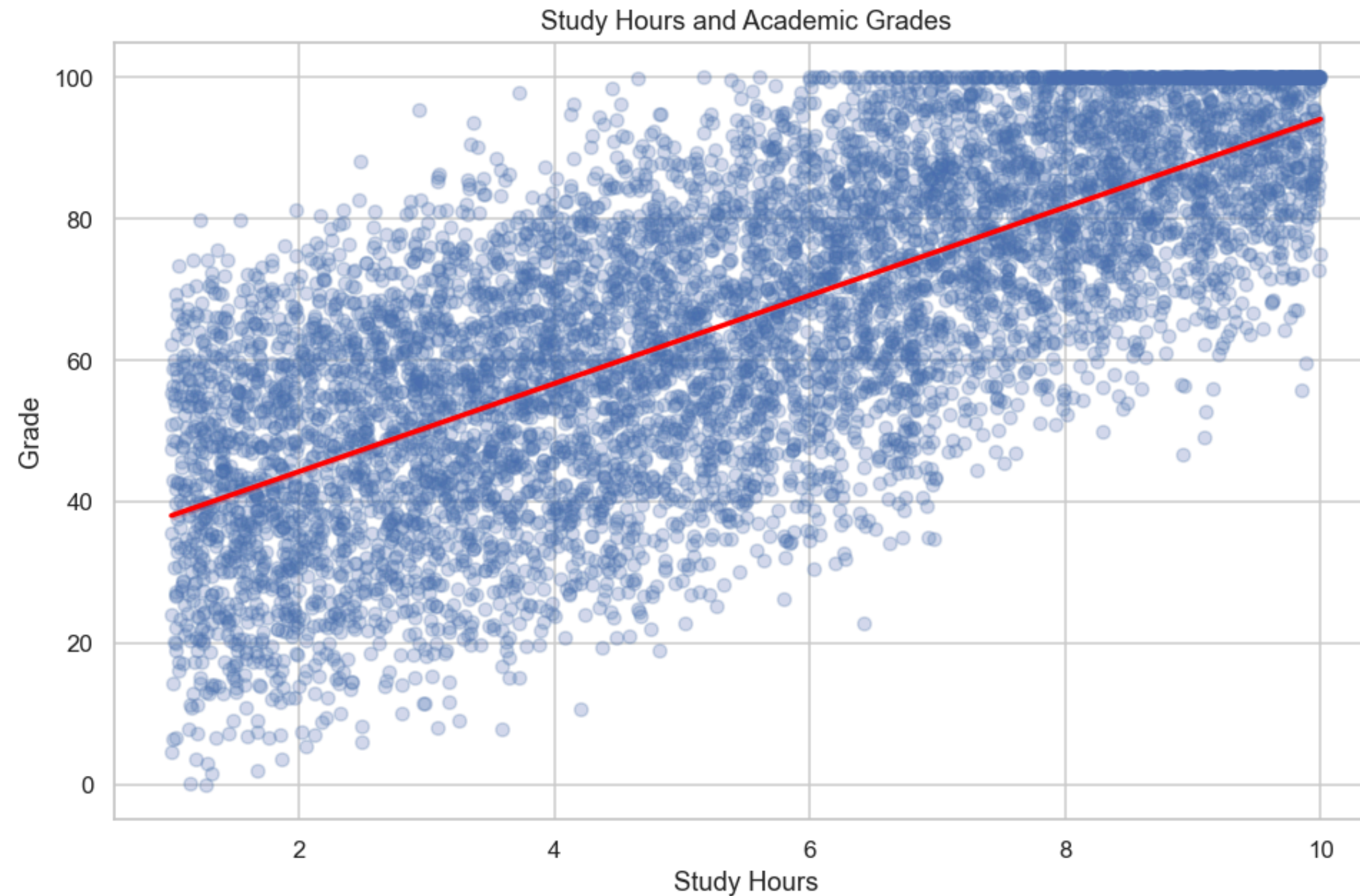
+30.7

POINTS, LOW TO HIGH STRESS

Mean grades rise from 49.84 (Low) to 72.18 (Medium) to 80.49 (High). Counterintuitive enough to read as a flag for validating how stress_level was recorded — not as evidence that stress helps.

FIGURE 3

Study Hours vs. Grades



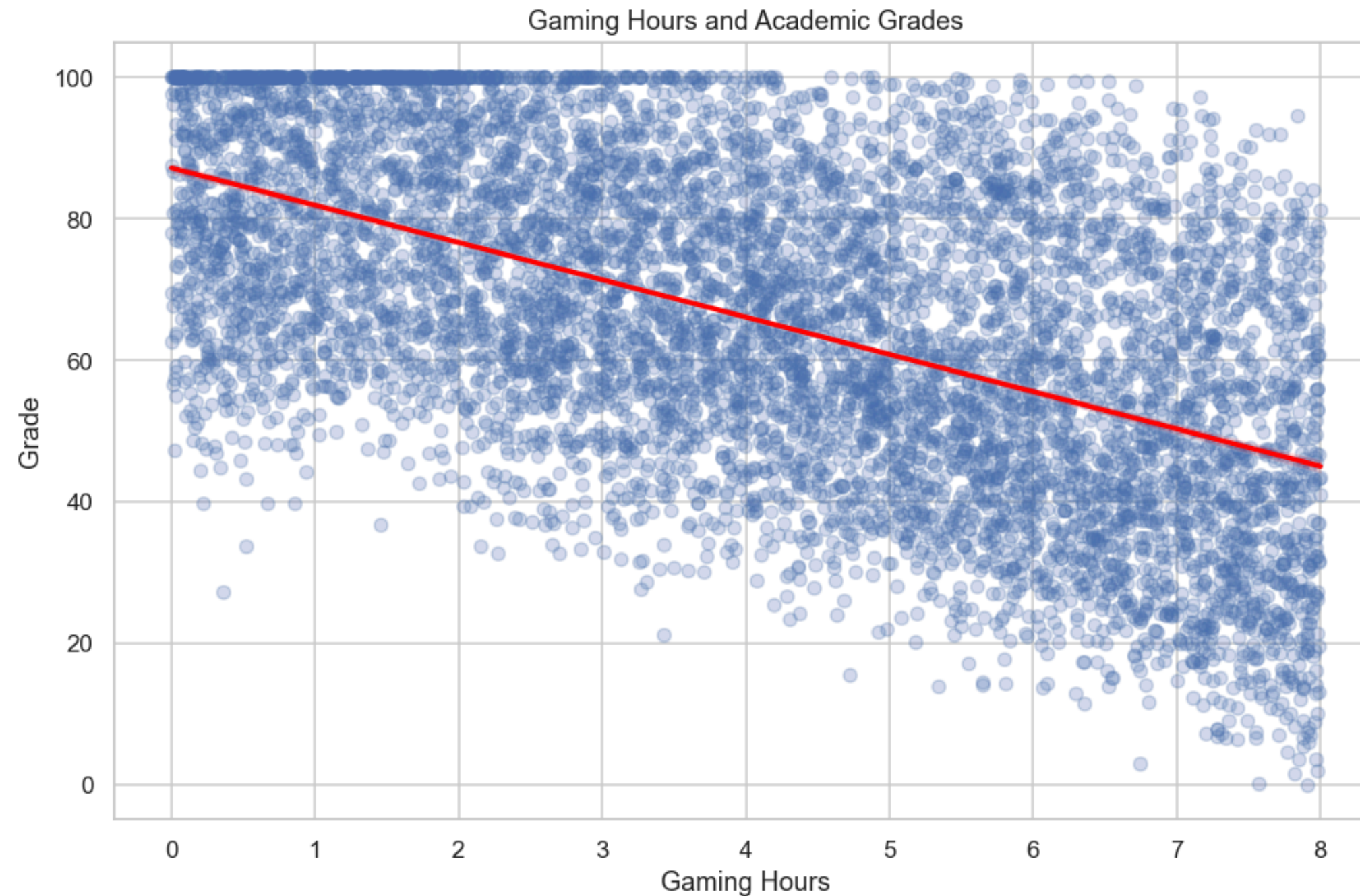
$$r = 0.725$$

STRONGEST POSITIVE
ASSOCIATION

Study time tracks grades more closely than any other variable here. The relationship holds across the full range, with wide scatter at every study level.

FIGURE 4

Gaming Hours vs. Grades



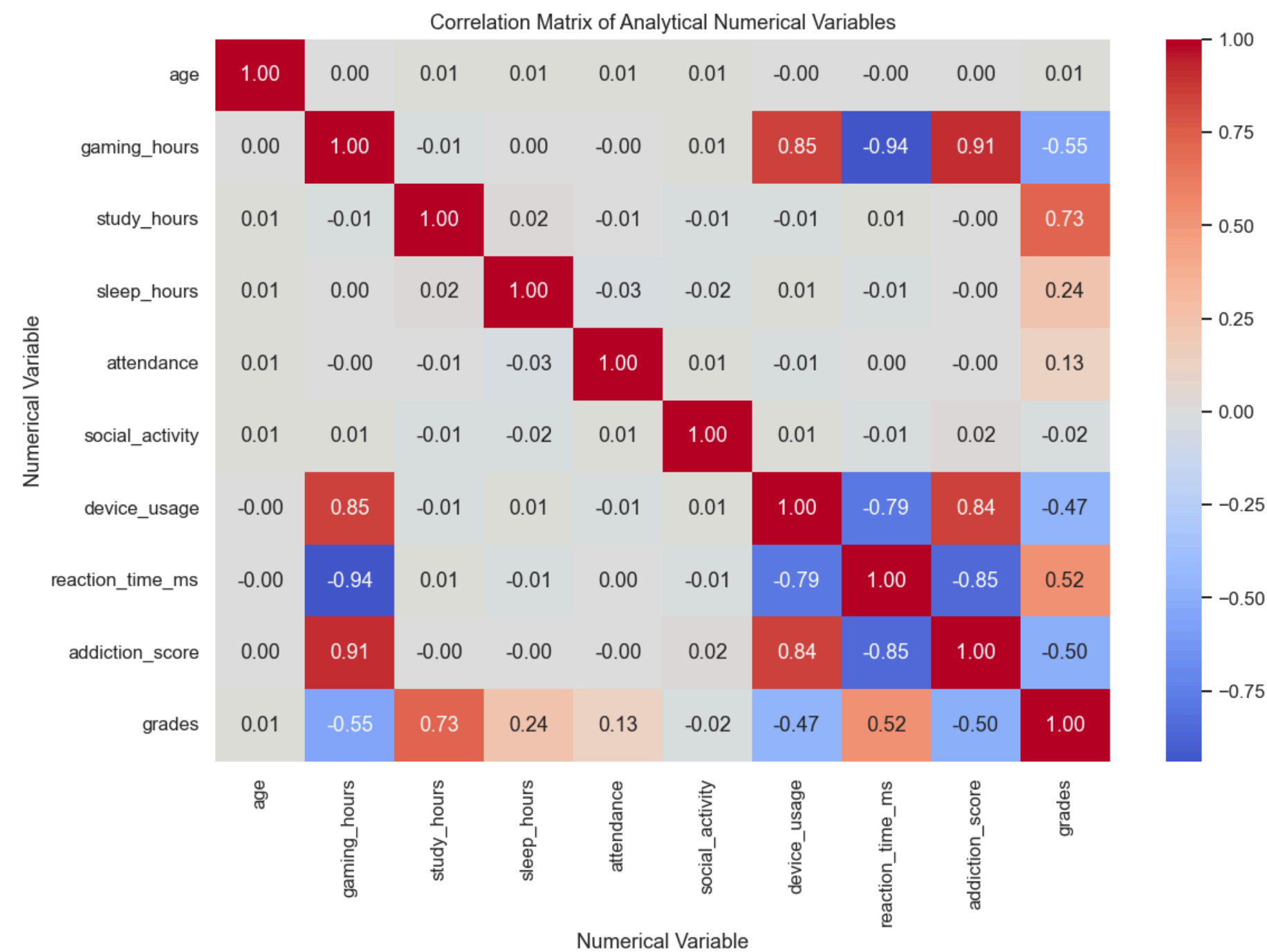
$$r = -0.554$$

MODERATE NEGATIVE
ASSOCIATION

Mean grades decline across gaming-hour bands. Weaker than the study-hours signal, and gaming hours do not travel alone — see the heatmap.

FIGURE 5

Correlation Heatmap



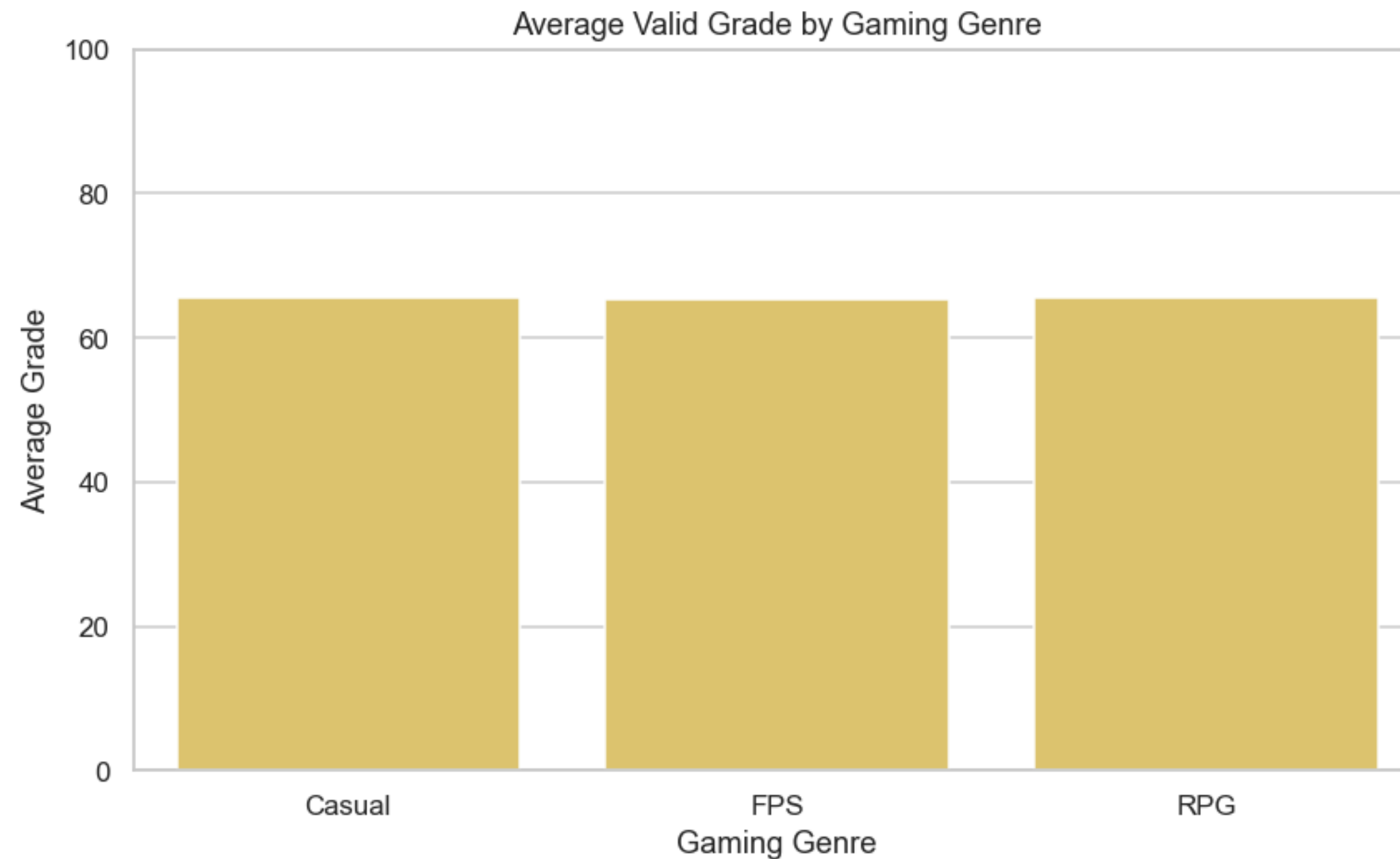
0.85–0.94

INTERCORRELATION BLOCK

Study hours (+0.725) and gaming hours (−0.554) are the strongest grade correlates. Gaming hours, device usage, addiction score and reaction time are so tightly intercorrelated that they cannot be read as independent effects.

FIGURE 6

Mean Grade by Gaming Genre



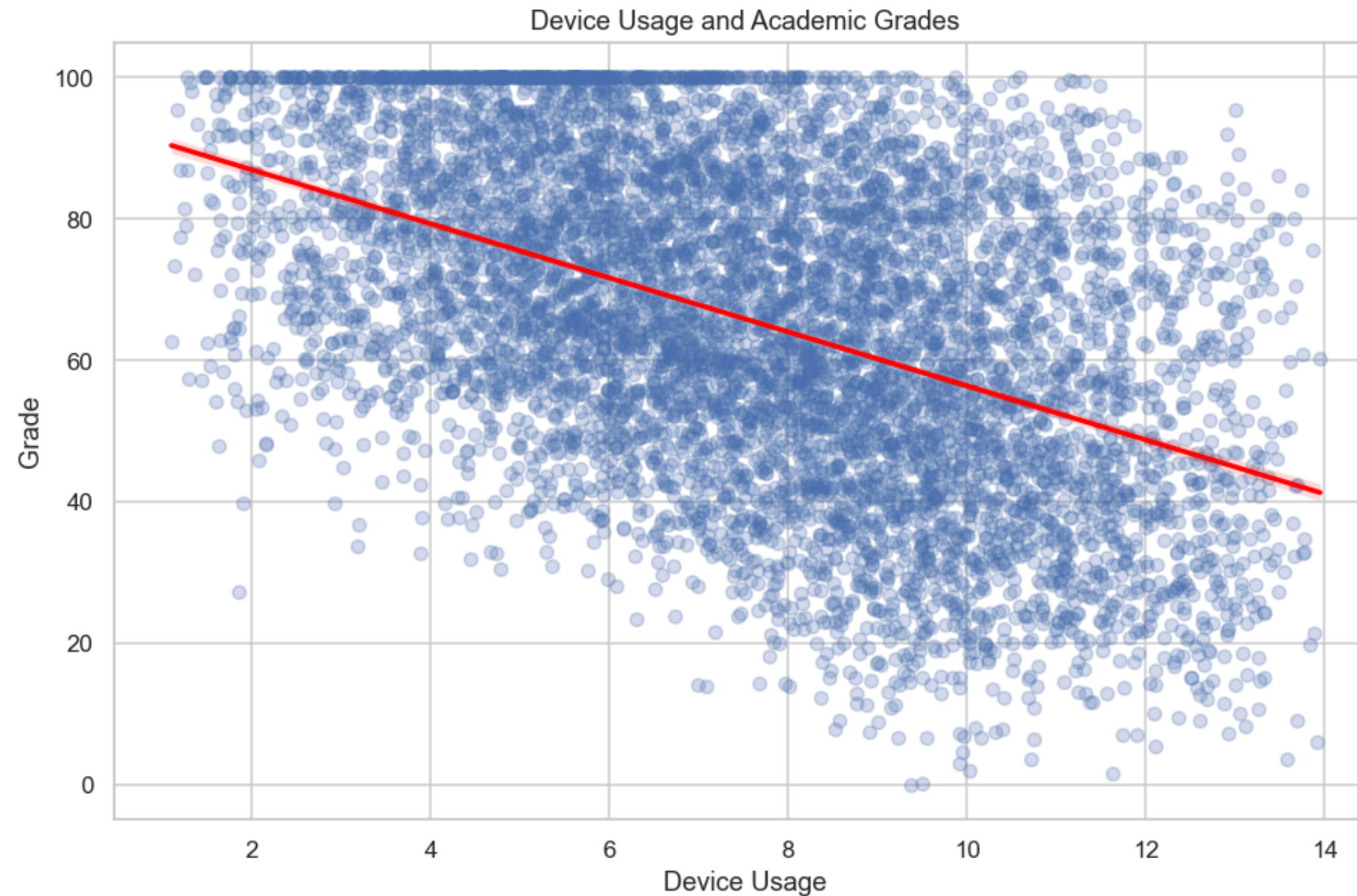
0.13 pt

SPREAD ACROSS ALL
THREE GENRES

RPG (65.60), Casual (65.55) and FPS (65.47) sit within a tenth of a point of each other. What is played carries no signal here; how long it is played does.

FIGURE 7

Device Usage vs. Grades



$$r = -0.470$$

MODERATE NEGATIVE
ASSOCIATION

Directionally consistent with gaming hours, but largely the same signal: the two correlate at 0.85 with each other. Treat them as one screen-time construct.

Key Insights

+0.725

Study hours are the strongest positive correlate of grades.

+0.238

Sleep is a weak positive signal, with attendance weaker still (+0.128).

+0.520

Reaction time runs counterintuitive its definition and provenance need checking.

231

Invalid values survived zero missing, zero duplicates and zero IQR outliers.

−0.554

Gaming hours show a moderate negative association.

−0.496

Addiction score and device usage (−0.470) are moderate negatives, but both are intercorrelated with gaming hours.

0.61 pt

Recommendations

Students

Balance study time with intentional limits on gaming.

Academic advisors

Screen for the combined low-study, high-gaming, high-device pattern rather than any single variable.

Educators

Treat sleep and attendance as secondary supportive signals, not primary levers.

Counselors

Use high gaming hours plus a high addiction score as a screening signal — never as a diagnosis.

Educators & advisors

Do not target interventions by gender or by gaming genre — the data supports neither.

Data team

Add validation rules rejecting grades above 100 and negative addiction scores; document the undocumented scales.

Research team

Move to longitudinal collection before any causal claim is entertained.

Reflection

HARDEST PROBLEM

Realizing that a dataset with no missing values, no duplicates and no IQR outliers still held 231 invalid ones.

TOUGHEST DECISION

Flagging and selectively nulling invalid cells instead of capping values or dropping whole rows.

MOST INTERESTING FINDING

The size of the study-hours effect, alongside the stress-level ordering that runs the wrong way.

DONE DIFFERENTLY

Secure the original source documentation, plan longitudinal collection, and preregister the validation rules.

THE EDA LESSON

Good EDA is measured by the quality of the evidence and the reasoning, not by the number of transformations. Correlation is still not causation.